

Robot that can mimic human walk

Scientists are close to developing a walking robot called AMBER 2 that could be useful for space exploration and elderly assistance. The team of scientists at the A&M Bipedal Experimental Robotics Lab, USA, led by Dr Aaron Ames, aims at making this bipedal robot mimic human walk.

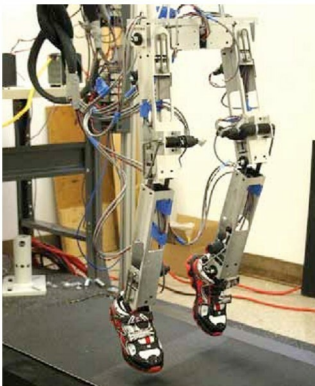
Informing about the project, Dr Ames said, "The first goal to overcome is how to mathematically model such a highly dynamic system, and then how to fabricate the controllers and other hardware to do what you want. Similar to what we chose to do with our first robot, we decided to create our next-generation AMBER 2 robot from the ground up." "The team realised it would need to use motors and gearheads in order to deliver torques between 8 and 12 Newton-metres and speeds down to 5 to 6 radians per second," he added.

The team is using three motors and three right-angle gearheads per leg—one for each joint (ankle, knee and hip)—and would use maxon motors for the final robot design.

Robot that learns language

The iCub humanoid robot will now be able to understand what is being said to it and even anticipate the end of a sentence. A team led by Peter Ford Dominey, CNRS Director of Research at Inserm Unit 846, France, has been working on this robot for many years.

This achievement was made possible by the development of a simplified artificial brain that reproduces certain types of so-called 'recurrent' connections observed in the human brain. The artificial brain system enables the robot to learn, and subsequently understand, new sentences containing a new grammatical structure. It can link two sen-



AMBER 2 robot that can walk like humans (Courtesy: <http://www.techclap.com/>)

tences together and even predict how a sentence will end before it is uttered. This research has been published in the journal PLoS One.

According to researchers, this advancement is of major importance to research into certain diseases, such as possible linguistic malfunctions in Parkinson's disease. This system can be used to understand better the way in which the brain processes language.

Foldable laptops coming soon

Coming one step closer to developing foldable laptops, researchers at the University of Maryland, USA, claim to have developed the world's flattest paper. The team printed transparent electronics onto a special kind of nanopaper that was made of wood pulp.

During the research, the scientists developed a transistor on the surface of the nanopaper by printing three different inks on it. Speaking about

the project in their research paper titled 'Highly Transparent and Flexible Nanopaper Transistor,' the authors wrote that the transistor had unique properties such as flexibility, cost efficiency, light weight and renewability. Also, the transistor can perform even when it is bent a bit as it is 84 per cent transparent and malleable.

Researcher Li-angbing Hu said, "The nanopaper is so thin that it can be cut or folded. It is as flat as plastic. The device configuration can be ap-

plied to many other semiconductor materials toward flexible green electronics."

Self-repairable electric wires

Researchers at the North Carolina State University have created electric wires that automatically reconnect together when damaged. According to the scientists, these stretchable wires are made of a liquid-metal core and a polymer sheath that "reconnect at the molecular level" when cut. Thus there would be very little need for maintenance when they are placed in high-stress environments.

Co-researcher Dr Michael Dickey said, "Because we are using liquid metal, these wires have excellent conductive properties. And because the wires are also elastic and self-healing, they have a lot of potential for use in technologies that could be exposed to high-stress environments."